

TraceMaker

Roadmap and test report

Every milestone, every test, every benchmark run, as of the close of the roadmap

Development report

6 October 2026

Abstract

This report records the state of the TraceMaker roadmap (milestones M0 to M13) and everything that was done to test the software: the test suite test by test, the build presets it runs under, the benchmark sets and the judge, every benchmark run on record, the held-out sets built for the placement gate, and each experiment of the close-out with its configuration and result. All fourteen milestones are closed: every task is either built and measured or closed with the measurement that shows why it was not built. Closed does not mean every gate is met. Two gates are not met (M8: from-scratch placement gives 85 % clean boards against a gate of 90 %; M9: 41 % of the dense-package boards are clean) and one comparison was never measured (M7 against the SA-PCB placer). The test suite has 132 tests and passes in the release and CPU-only builds; the address-sanitizer build reports no memory error. On the main routing benchmark the final binary routes 100, 70.0, 63.3 and 54.5 % of the sampled boards of tiers A to D with no error added by routing, against 100, 50.0, 46.7 and 36.4 % published for Freerouting 2.5.

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1 How to read this report

The tables in Sections 11, A, B and C, and the summary tables that carry counts, are generated by `report/make_test_report.py` from the project’s own records: `dev/progress.json` (the roadmap), the JUnit file of a full test run, the test sources, and `bench/results/*/summary.json` with their per-board files. A number in those tables is the number in the record. The text sections were written by hand from the design documents (`docs/`), the decision log (`docs/12-decisions.md`, D1 to D64) and the experiment logs; each experiment names the files that hold its raw results.

Three words are used with fixed meanings throughout.

Clean. A routed board is clean when KiCad’s own design-rule check reports no unconnected item and no error that the run added (Section 3.1).

Closed. A roadmap task is closed when it is built and measured, or when a measurement shows that building it would not help and it is recorded as not built with that measurement.

Gate. The test or benchmark result a milestone was meant to reach. A closed milestone can have an unmet gate; Table 2 says which.

2 Roadmap summary

	Milestone	Tasks	Built	Closed	unbuilt	Gate
M0	Toolchain and skeleton	14	14		0	met
M1	KiCad I/O and board model	11	11		0	met
M2	Rules and DRC parity	11	11		0	see text
M3	Event bus, server, viewer v1	9	8		1	not measured
M4	First router (lattice A*)	6	6		0	met
M5	Failure memory T0–T2 + escalation R0–R3 + repair	10	10		0	met
M6	Global routing + GPU	10	8		2	met
M7	Placement A–D	12	12		0	not measured
M8	Placement E–G + ECO	12	8		4	not met
M9	Escape planning	9	7		2	not met
M10	Portfolio, determinism, knowledge base	10	10		0	met
M11	KiCad integration	8	8		0	met
M12	Advanced rules and arms	12	10		2	met
M13	Component-aware layout rules	17	17		0	met

Table 1: The fourteen milestones. “Built” counts tasks that were implemented and measured; “closed unbuilt” counts tasks closed with a measurement instead of an implementation. The gate column is “met” when the progress record carries no dissenting gate result.

	Gate	Result
M0	CPU and CUDA tests on both GPUs; <code>kicad-cli</code> works from the harness	met
M1	Byte-identical round trip of every fixture board; netlist equals KiCad's	met
M2	Same violations as <code>kicad-cli pcb drc</code> on all fixture boards	14 of 17 demo boards exact, 264 of 264 injected-defect boards; three boards differ in documented ways
M3	Viewer at 60 frames per second; replay reproduces the board	replay met ; frame rate not measured (no GPU-backed browser on the machine)
M4	Simple set routed with zero added KiCad errors	met
M5	Completion at least Freerouting's on its test set	met (tier A: 100 % clean, equal to Freerouting 2.5)
M6	CPU and GPU identical; time to route reduced on dense boards	met for the GPU workload in use (cost-to-go fields); global routing did not contribute
M7	Legal placements; wirelength at most SA-PCB's	legality met ; the SA-PCB comparison was not measured (the baseline does not build without SWIG)
M8	From-scratch clean pass at least 90 %; <code>eco</code> closes boards	not met : 85 % (34 of 40 held-out boards)
M9	BGA boards complete	not met : 41 % of 17 dense-package boards clean (Freerouting 2.5: 24 %)
M10	Bit-identical across thread counts and GPU on/off	met
M11	Round trip in KiCad on the demo boards	met , in a running KiCad 10.0.6 GUI driven without a desktop
M12	Each feature gated by its own subset; no regressions	met per feature (Section 8)
M13	Detection precision and recall; every applied rule met or reported; no regression	met

Table 2: Gates and their results.

3 How TraceMaker is tested

3.1 The judge

Every routing and placement result is judged by KiCad itself, not by TraceMaker's own checker: `kicad-cli pcb drc --severity-all --all-track-errors` from the Docker image `kicad/kicad:10.0.6`, run on the input board and on the output board (`bench/run.py`, function `drc`).

- **Added routing errors** are the violations of severity *error* that involve a track, an arc or a via, counted by type, output minus input. Types that routing cannot cause (silkscreen, courtyards, library mismatches, unconnected items, dangling tracks) are excluded from this count; unconnected items are counted separately.
- **Added placement errors** (when TraceMaker moved parts) are courtyard overlaps, holes inside courtyards, copper-to-edge and copper clearance, shorts, solder-mask bridges and hole clearances between different footprints, placed board minus the designer's board.
- A board is **clean** when KiCad lists no unconnected item on the output and both counts above are empty.
- **Completion** is one minus unconnected items after over unconnected items before.

PCBench boards come without their project files, so KiCad's default rules apply (track 0.25 mm, clearance 0.2 mm, via 0.8/0.4 mm). Section 5.3 shows what that does to the benchmark: only 6 of 92 designer-routed boards pass the same judge.

3.2 Environment

Machine	56 cores, about 150 GB RAM, Tesla P100 (12 GB) and Tesla V100 (16 GB), shared with other jobs
Compilers	GCC 15 for C++20; CUDA 12.4 with g++-13 as host compiler
Warnings	-Wall -Wextra -Wpedantic -Wshadow -Wconversion and more, with -Werror
KiCad	10.0.6 from Docker (kicad-cli, and pcbnew for the GUI test)
Freerouting	published PCBench results for 2.5.0-RC12 and 2.4.1; 2.5.0-RC12 and 1.9.0 run locally for the quality study
Test framework	Catch2 3 through CTest, JUnit results per test

Table 3: Test environment.

Four build presets are tested. **release** is the one benchmarks use. **cpu-only** builds without CUDA and proves that everything works with no GPU (GPU tests skip). **asan** adds the address and undefined-behaviour sanitizers. **tsan** (thread sanitizer) exists as a preset and was not run in the close-out.

3.3 Levels of testing

1. **Unit tests** (Catch2, 116 test cases in 15 source files): geometry predicates, the s-expression parser and writer, the KiCad reader, rule reading, the random-number generator, the placer’s parts, the component-rule catalogue and detection, impedance formulas against textbook and KiCad values, the event server and the replay log.
2. **Reference-path equivalence** (non-negotiable rule 3 of the project): every accelerated path has a plain reference path and a test that both give the same result. Table 4 lists them.
3. **Integration tests** (16, scripts under `tests/integration`): they run the real binaries on real boards and compare with KiCad, with a second run, or with the reference path.
4. **Benchmarks** judged by KiCad on PCBench tiers A to D, the dense-package set, DAC 2020, and a held-out quality study against Freerouting run on the same machine.
5. **Held-out sets** for placement: H (40 boards never used before) and S (the same boards with their parts piled), built for the M8 gate (Section 6).
6. **A running KiCad**: the plugin against `pcbnew` under a virtual display (Section 7).

Accelerated path	Reference path	Test
GPU Philox random fill	CPU fill	bit-identical on every device (<code>test_gpu.cpp</code>)
GPU cost-to-go fields (CUDA sweeps)	CPU sweeps	equal on random grids (<code>test_gpu.cpp</code>); routing identical with and without GPU (<code>thread_determinism</code>)
Zone-fill edge index for the DRC	every edge of every fill (<code>drc --linear-zones</code>)	[<code>geom</code>] equivalence test on random shapes; <code>drc_zone_index</code> on four boards; vme-wren and jetson compared by hand, byte-identical
Polygon index for disks	<code>closer_than</code>	[<code>geom</code>] test, 4,000 disks
Flood-fill reachability pre-check	full A*	<code>reach_verify</code> : every “unreachable” re-checked, zero mismatches
Incremental routability term in the annealer	from-scratch recompute	placer unit test
Portfolio on a thread pool	one thread	<code>thread_determinism</code> : identical bytes at 1, 3 and 8 threads
Placer’s translated closeness test	<code>geom::closer_than</code>	placer unit test
Exact windows by branch and bound	full enumeration	placer unit test
Python module	command-line binary	<code>python_bindings</code> : identical boards
Recording and live view	plain routing	<code>record_replay</code> : identical boards

Table 4: Accelerated paths, their reference paths and the tests that hold them equal.

3.4 Determinism as a test method

The router has two budgets. `--time` is a wall-clock limit and makes results depend on machine load. `--work` is a budget in search expansions: the same input, seed and budget give the same bytes at any thread count and with or without a GPU (decision D47). Every experiment in Section 8 was first measured with `--work` (one variant or all eight), where a difference of one connection is real, and only then on the KiCad-judged tiers at 120 seconds, where the eight-variant portfolio and machine load add noise. Two results in this report show why both are needed: a planning change that routed 81 more connections at a fixed budget was neutral on the tiers, and a tier run made while two others were running read 52 connections lower than the same binary side by side with its control.

3.5 Rules for experiments

- Compare two configurations of the *same binary*, run at the same time on the same machine.
- Report losses and null results with the same weight as gains; a change that does not help is removed or left off, and the decision log says so.
- Never tune on held-out boards. Sets H and S were built from boards that appear in no earlier result, list, test or document.
- The quick tier (A) is run before a change to routing or placement is finished.

4 The test suite

4.1 Summary by source file

Source	Tests	Passed	Skipped	Failed	Time (s)
src/crules/test_crules.cpp	19	19	0	0	0.27
src/crules/test_impedance.cpp	16	16	0	0	0.11
src/place/test_place.cpp	28	28	0	0	30.39
src/server/test_replay.cpp	5	5	0	0	5.20
src/server/test_server.cpp	4	4	0	0	0.37
tests/test_diff_pair.cpp	5	5	0	0	0.02
tests/test_escape.cpp	5	5	0	0	3.14
tests/test_geom.cpp	7	7	0	0	0.63
tests/test_gpu.cpp	4	4	0	0	1.99
tests/test_kicad_io.cpp	8	8	0	0	0.42
tests/test_netlist.cpp	1	1	0	0	0.01
tests/test_rng.cpp	2	2	0	0	0.01
tests/test_rules.cpp	4	4	0	0	0.02
tests/test_sexpr.cpp	7	7	0	0	0.03
tests/test_units.cpp	1	1	0	0	0.00
integration (tests/CMakeLists.txt)	16	16	0	0	261.57
132 tests: 132 passed, 0 skipped, 0 failed; 304 s of test time					

Table 5: Release build, full run at the end of the close-out. Appendix A lists every test.

4.2 Results by build preset

Preset	Result at the end of the close-out
release	132 of 132 pass (303 to 440 seconds depending on load).
cpu-only	132 of 132: 130 pass, the two GPU equivalence tests skip as designed.
asan	No sanitizer report in any test. Unit tests pass. The slow integration tests were re-run one at a time on an idle machine after their limits were adjusted (below) and pass: drc_zone_index 546 s, global_route 526 s, diff_pair 891 s, thread_determinism 1,163 s, crules_two_stage 1,654 s. The preset was not rerun end to end afterwards.
tsan	Not run in the close-out.

Table 6: Test results per preset.

What the sanitizer run exposed in the tests themselves. On a fully loaded machine four integration tests ran into their time limits under **asan** and **diff_pair** failed with 111 of 113 connections routed. None was a memory error. **diff_pair** set a work budget but left the router’s default 120-second wall clock in force, so a slow build stopped early; it now passes `--time 3600`, which makes the work budget the only limit. **crules_two_stage** anneals at a fifth of the default effort (it checks legality and locking, not quality). The two tests added in the close-out were made lighter (four boards instead of six; a 3 M instead of an 8 M budget), and the limits of the slow tests were raised to 30 and 60 minutes.

4.3 What the integration tests check

Appendix A carries a one-line description under each of the sixteen integration tests. The three that compare with KiCad carry the most weight:

- **kicad_drc_parity**: TraceMaker’s DRC against KiCad’s on 14 demo boards, violation types equal.
- **kicad_drc_broken_parity**: 264 boards with injected defects, equal violation by violation.
- **kicad_edit_roundtrip**: boards written by TraceMaker read back identically in pcbnew.

Untouched content of a `.kicad_pcb` file round-trips byte for byte (rule 8); that is tested on the fixture boards in `test_kicad_io.cpp`.

5 Routing benchmarks

5.1 Sets

Set	Source	Boards	Use
PCBench tier A	Freerouting’s bench-mark fixtures	40	quick tier: run before any routing change is finished
PCBench tier B	same	40	main comparison with Freerouting’s published results
PCBench tier C	same	30	same
PCBench tier D	same	22	largest boards (600 to 1,600 connections)
Dense-package set	PCBench boards with BGAs and fine-pitch parts	17	escape planning (M9)
Quality study	60 PCBench boards unused in development	60	Freerouting 2.5.0-RC12 and 1.9.0 run locally, same judge
Hard set	10 to 24 large and nearly clean boards	24	fixed-budget experiments (<code>bench/m6_bench.py</code>)
KiCad demos	KiCad 10	17	DRC parity, rule coverage, the GUI test
DAC 2020	hand-routed boards	11	vias and layers against human routing
H / S	held-out, built 2026-10-06	40	placement gate (Section 6)

Table 7: Benchmark sets. Tier samples are seeded and fixed; the same boards are used in every tier run.

5.2 Final result

All tier runs use eight router variants on eight threads, 120 seconds per board, and KiCad’s DRC as judge. Table 8 compares the binary at the start of the close-out with the final one. The final four tiers were run at the same time (64 router threads on 56 cores), so their routed totals are a little low; the clean-pass counts are what the table is for.

Tier	Start of close-out (reach1)		Final (final11)		Freerouting	Boards with added errors
	Clean	Routed	Clean	Routed	2.5 clean	
A (40 boards)	100 %	2,637	100 % (40)	2,637	100 %	0
B (40)	67.5 %	5,165	70.0 % (28)	5,172	50.0 %	0
C (30)	63.3 %	9,523	63.3 % (19)	9,516	46.7 %	0
D (22)	50.0 %	16,628	54.5 % (12)	16,522	36.4 %	0

Table 8: Clean pass and connections routed. Tier D’s twelfth clean board (m2fc) closes its last connection or not from run to run; side by side earlier the same day tier D read 16,618 and 16,622 routed with 12 clean both ways. Tier B’s gain is one board from the half-pitch variant (Section 8).

Appendix B lists every board of the final runs; Appendix C lists every benchmark run on record since the project began.

5.3 What the benchmark measures, and what it cannot

Three studies in the close-out examined the benchmark itself.

The designers’ own layouts under the same judge (`bench/human_baseline.py`; the routed `raw.kicad_pcb` of the 92 boards of tiers B to D).

Designer’s layout clean	6 of 92
TraceMaker non-clean boards whose designer’s layout is not clean either	33 of 34
TraceMaker clean boards whose designer’s layout has errors on routed copper	52 of 58 (median 108 errors)
Designer used copper zones that the unrouted fixture no longer has	82 of 92
Open connections on non-clean boards that are on a net the designer poured	32 of 938

Table 9: The benchmark is judged under rules that the original designs mostly do not meet.

What if the poured nets were left to a pour. The 34 non-clean boards were routed twice at the same fixed budget (eight variants, 60M expansions): with all nets, and with the designer’s zone nets skipped. Four boards complete without them; open connections fall from 938 to 723. Missing pours contribute on a few boards and are not the main cause.

Why the last connections fail. After a reporting fault was fixed (Section 9), nine nearly clean boards were examined: on five, the last one to five connections are lost to nets that keep ripping each other at one package; one board has pins sealed by fixed copper; two are boxed in before negotiation. On `serial_gw` the geometry is visible: a QFN at 0.5mm pitch whose centre pad has no net, pins 3 to 6 alternating ground and supply, and no room for the vias the crossing needs under the default rules.

6 Held-out placement sets and the M8 gate

Set H (`bench/place_sets/H.txt`, built by `bench/make_place_sets.py`, frozen). A seeded sample (seed 7) of PCBench boards that appear in no earlier result, development list, test or document (342 boards excluded), with 10 to 300 parts and at least 10 movable, a courtyard on at least 80 % of the parts, and routed completely in the designer’s placement in 120 seconds. Twenty boards at or below the candidates’ median of 33 parts and twenty above: 14 to 138 parts, 4,022 connections; 132 candidates were tried.

Set S. The H boards with every part that full-mode placement may move piled at the centre of the board, so the file no longer contains the designer’s placement. Parts the placer holds in the designer’s board (locked parts, edge connectors, mounting holes, parts overhanging the edge) stay, since a schematic does not say where a connector goes.

Run	Clean	Routed
H, designer's placement, before the two router fixes (H- none-1)	38 of 40	4,021 of 4,022
H, designer's placement, after the Margin fix (H- none-2)	38 of 40	4,019 of 4,022
S, full mode as the placer was (S- full-1)	28 of 40 (70 %)	3,900 of 4,002
S, full --scratch on the 12 failing boards (S- full-4)	3 of 12	
S, routable --scratch, first version (S- routable-5)	33 of 40 (82.5 %)	3,944 of 4,022
S, routable --scratch, final (S- routable-7)	34 of 40 (85 %)	3,939 of 4,013

Table 10: The M8 gate asks for 90 %. It is not met.

The six boards that fail in the final run: three have a part that fits nowhere without breaking copper, hole or mask rules that the designer's placement itself breaks (an Arduino module; a 15 mm IC between two pin headers 12.4 mm apart; a battery holder); kitspace_hack cannot be routed clean in the designer's placement either once Margin lines count as edges (89 of 92); ottawa-badges-2016 and pico-pi are three connections short each. Of the 36 boards that can be clean, 34 are. More spacing between parts was tried on the two near misses (courtyard clearance 0.6 and 1.0 mm) and routed fewer connections (211 and 191 of 219 on pico-pi). Placement takes 91 seconds at the median in routable mode. Table 11 gives every board.

Table 11: Held-out boards: the designer's placement (H) and from-scratch placement (S, final run). "pl. err." are KiCad errors added by placement.

Board	Parts	Movable	H: routed, judged	S: routed, judged	Place (s)
2d_conduction_sk9822-matrix	133	122	374/374 clean	374/374 clean	82
4_digit_hex_display_4_digit_display	46	44	108/108 clean	108/108 clean	334
8088_sbc2_pcb_8088_sbc2	65	60	328/328 clean	328/328 clean	583
ABOVISP_ABOVISIP	15	11	31/31 clean	31/31 clean	183
avr_frs_shield_avr_frs	34	30	62/62 clean	62/62 clean	27
bmw-ibus-bluetooth_bmw_bt_cdcemu_digital	58	50	125/125 clean	125/125 clean	143
Box0-hv-analog-breakoutboard_breakout	57	54	149/149 clean	149/149 clean	338
bypass_crossmix_bypass_crossmix	61	53	122/122 clean	122/122 clean	108
clunet-switch1_switch	36	24	69/69 clean	69/69 clean	32
custom_cpu-register_custom_cpu-register	47	20	74/74 clean	74/74 clean	36
digital_clock_led_clock_v1	138	133	251/251 clean	251/251 clean	590
fuzzer_fuzzer	19	10	69/69 clean	69/69 clean	39
i2lcd__autosave-i2lcd	22	22	74/74 clean	74/74 clean	264
Kefersender_UKW TX	40	36	84/84 not clean	84/84 clean	68
kitspace_beehive	29	15	79/79 clean	79/79 clean	46
kitspace_GPSMux	20	12	36/36 clean	36/36 clean	18
kitspace_hack	29	24	89/92 not clean	84/92 not clean	299
kitspace_LED%20Zappelin	42	37	118/118 clean	118/118 clean	44
kitspace_postcard	20	20	37/37 clean	37/37 clean	20
kitspace_tt_nano_HAT_b2	25	20	56/56 clean	30/56 not clean	328
LiPoSensor_lipo_sensor	31	26	72/72 clean	72/72 clean	51
MAVRIC_Hardware_SoilBoard	15	13	39/39 clean	39/39 clean	116
medusa_medusa_rs422_rx	46	44	97/97 clean	97/97 clean	83
Mini-Ultra-8-MHz_RS-MINI-ULTRA-8M	22	17	67/67 clean	32/61 not clean (3 pl. err.)	276
NoVo-Pi_NoVoPi	24	23	44/44 clean	44/44 clean	16
NRC2016_usb_sio	26	25	83/83 clean	83/83 clean	91
Open-Source-Power-Supply_PowerSupply_PCB_backup	54	46	90/90 clean	90/90 clean	77
ottawa-badges-2016_ottawa-badges-2016	14	13	30/30 clean	22/27 not clean (7 pl. err.)	193
ottawa-badges-2016_ottawa-badges-2016	98	86	200/200 clean	197/200 not clean	645
pcb-covox-amp-v2_pcb-covox-amp-v2	61	56	125/125 clean	125/125 clean	77
Phased-Array-Microphone-using-FPGA_SateliteMicrophone	21	17	33/33 clean	33/33 clean	90
photon_Sprinkler_sprinkler	17	16	39/39 clean	39/39 clean	17
Pi-Zero-SD-and-Audio-Out_BaseBoard	57	45	146/146 clean	146/146 clean	62
pico-pi-rel_pico-pi	96	82	219/219 clean	216/219 not clean	414
PiPlay_AudioHAT	43	33	108/108 clean	108/108 clean	50
PowerGloveUHID_sensor_board	18	14	28/28 clean	28/28 clean	132

Table 11: Held-out boards: the designer’s placement (H) and from-scratch placement (S, final run). “pl. err.” are KiCad errors added by placement.

Board	Parts	Movable	H: routed, judged	S: routed, judged	Place (s)
RGBMatrixPanelCPLD-PhotonBackpack_	25	19	85/85 clean	85/85 clean	106
RGBMatrixPanel_CPLD_negative					
sensorboard_DiffIR	19	15	32/32 clean	32/32 clean	125
TLPHPnodeV2_TLPHPnodeV2	34	32	89/89 clean	89/89 clean	304
z2amiller_sensorboard	25	22	58/58 clean	58/58 clean	23

7 KiCad in the loop without a desktop

KiCad 10’s programming interface only works against a running `pcbnew`. The script `scripts/kicad_gui_roundtrip.py` starts a virtual display (Xvfb), runs `pcbnew` from the Docker image on it with the API server enabled, dismisses KiCad’s first-run wizard by synthetic mouse clicks, and then drives TraceMaker’s plugin against the live board through the API socket. It checks, in order:

Check	Result on <code>multichannel_mixer-unrouted</code>
KiCad starts and opens its API socket	yes
The API answers after the wizard is dismissed	yes (KiCad 10.0.6)
The plugin runs against the live board	80 connections in 30 seconds
Copper appears in the open board	tracks 140 → 316, vias 19 → 21
One undo removes the whole result	140 tracks, 19 vias
Redo brings it back	316 tracks, 21 vias
KiCad’s DRC of the board saved from the GUI	unconnected 148 → 65, no new errors
With placement (<code>--place refine</code>)	65 footprints moved; old copper routed again; zones refilled
Two undos (zone refill, then the commit) restore everything	140 tracks, 19 vias, footprints back
KiCad’s DRC with placement	unconnected 148 → 101, no new errors

Table 12: The plugin in a running KiCad. The script is not part of the automatic suite: it needs Docker, Xvfb and about three minutes.

The plugin package was also installed through KiCad’s Plugin and Content Manager in the same headless session (Figure 1): the manager lists TraceMaker 0.1.0 as installed and compatible, and the files are in KiCad’s third-party plugin folder. Not checked: that KiCad then builds the plugin’s Python environment and shows its toolbar button, because the container had no network access for `pip`.

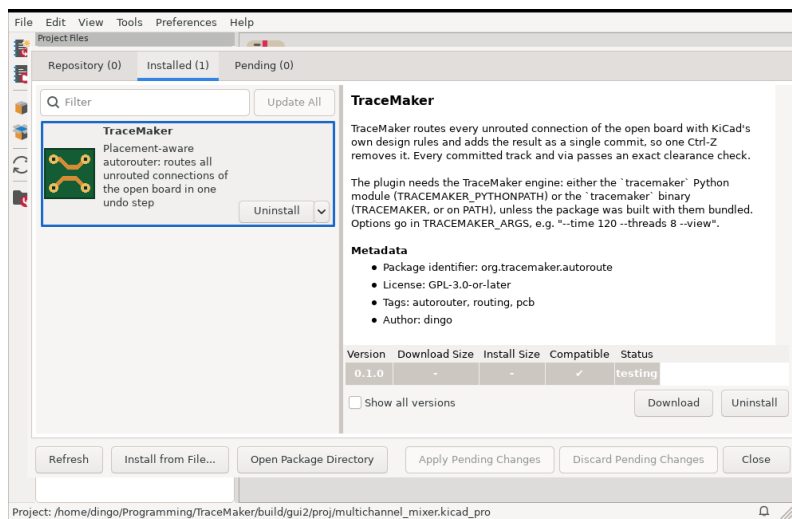


Figure 1: KiCad’s Plugin and Content Manager after “Install from File”, captured from the virtual display.

8 Experiments of the close-out

Each row is one experiment: what was compared, on what, and the result. “Fixed budget” means `--work`, so the numbers are repeatable to the connection.

Table 13: Experiments, 6 October 2026. Raw results are under `build/` and `bench/results/` in the files named; the design documents give the full account (`docs/05-routing.md` §16 to §25, `docs/04-placement.md` §10).

Experiment	Configuration	Result	Outcome
Global congestion map	18 hard boards, one variant, 100 M expansions; map at three settings against none (<code>build/m6-cong2.txt</code>)	13,285 routed without; 13,289 to 13,294 with; corridors 12,933	off (D53)
Tile capacity model	boundary samples against the narrowest of eight cut lines	tiles at or above 75 % full on Aleste: 1.7 % $\rightarrow$ 4.3 %	kept
Group re-route of conflicting nets	up to five connections, every order; 8 nearly clean boards (all variants, 60 M) and 24 boards (one variant)	identical routed count on every board	removed
Planning by copper distance	24 boards, one variant, 60 M (<code>build/m8-r4.txt</code>); then tiers	11,905 $\rightarrow$ 11,986 routed (15 boards better, 4 worse); tiers A to D neutral, tier D side by side 16,618 $\rightarrow$ 16,622	on (D54)
DRC profile on large zone boards	vme-wren (38,062 copper items, 128 zones), per-phase timing	broad-phase 14 ms; exact tests against zone fills 650 of 730 s	index built
Zone-fill edge index	vme-wren and jetson, indexed against linear path	730 s $\rightarrow$ 7 s and 250 s $\rightarrow$ 4 s; reports byte-identical	on (D55)
Global router v2	18 hard boards, one variant, 100 M (<code>build/m6-v2.txt</code>)	13,305 without; 12,967 corridors; 12,735 confined; 13,324 map	off (D56)
Global routing time	per board, timing output	tile-graph routing 0.02 to 0.5 s; set-up 0.4 to 1.3 s	no GPU port
Cut-line proofs (R6)	92 boards of tiers B to D (<code>build/cut-report-tiers.json</code>)	no over-full line; tightest line 9 % used at the median, 40 % at most	built, report only
Successive halving	24 boards, 30 M per thread, 2 and 4 threads (<code>build/m10-halving.txt</code>)	12,385 and 12,451 against 12,402 and 12,508 for the first variants	off (D59)
Micro vias	two four-layer boards with the setting forced on, 20 M	routing complete or unchanged; KiCad clean; none needed	built, off
Half the lattice pitch	12 nearly clean boards, all variants, 30 M against 120 M at half pitch (<code>build/m12-pitch.txt</code>)	4,140 $\rightarrow$ 4,153 routed; one more board complete; 9 better, 2 worse	one variant
Half-pitch variant on the tiers	tiers B and C, with and without, side by side (<code>ab-fine-*</code> , <code>ab-nofine-*</code>)	B 27 $\rightarrow$ 28 clean, 5,156 $\rightarrow$ 5,163; C 18 $\rightarrow$ 18, 9,506 $\rightarrow$ 9,514; no added errors	on (D63)
Pair twists	18 boards, 59 pairs, one variant, 30 M	one pair 0 $\rightarrow$ 86 % coupled, none worse; at least half coupled 35 $\rightarrow$ 36; median share 57 $\rightarrow$ 63 %; routed 2,210 $\rightarrow$ 2,212	on (D64)
Why pairs fail	same boards, tagged first attempts	63 of 120 attempts couple; 20 of 26 “mirrored” failures are one pair retried; 13 of 21 attempts joining routed copper couple	measured
Net classes from rules	StickHub, CM5 Minima, a unit test on a synthetic four-layer board	USB pair 0.760/0.150 mm for 90 Ω ; supply 1.367 mm for 3 A; exactly the two nets resolve to the class	built, opt-in (D58)
Part spacing on near misses	two S boards, courtyard clearance 0.6 and 1.0 mm	fewer connections routed (pico-pi 216 $\rightarrow$ 211 and 191)	not adopted

9 Faults found by testing

Fault	How it was found	Fix and its test
Copper routed across Margin-layer lines	Held-out board kitspace_hack: KiCad reported 13 to 24 edge-clearance errors on new tracks, with the designer's placement too	Margin graphics are edges in the router and in TraceMaker's DRC; DRC parity tests; re-route adds no edge error
Hole-less pads connected on the far side	Held-out board Kefersender: the router reported 84 of 84, KiCad 2 unconnected	A pad without a hole has copper on its footprint's side only; KiCad then agrees (0 unconnected)
Failure reasons were stale	Reasons disagreed with what a single-board run showed	Reasons are saved with the best state; ripped connections name the net that ripped them
DRC took minutes on boards with large zones	Timing vme-wren for the broad-phase item	Edge index and rule cache, with the linear path kept as reference
Parts left stacked by from-scratch placement	Set S: shorts and courtyard overlaps in KiCad	--scratch; fallbacks never return to the input
Cut "proof" on MicroMaple was wrong	The proof contradicted routed results	Demand counts only pads wholly on either side with no pad across the line
Pair lost its coupling after a planning change	diff_pair test failed	Runs with pair routing keep the old planning
A work-budget test depended on wall time	Sanitizer run on a loaded machine	Test passes --time 3600

Table 14: Faults found during the close-out. The first two were present in every earlier benchmark run.

10 What is not tested or not measured

- **M7's comparison with SA-PCB.** The baseline placer needs SWIG for its KiCad parser and its plain Makefile is stale; it was cloned (build/tools/sa-pcb) and not built.
- **The viewer's frame rate.** No GPU-backed browser is available on the machine.
- **The thread-sanitizer preset** was not run in the close-out, and the address-sanitizer preset was not rerun end to end after its slow tests were adjusted.
- **The plugin's toolbar button in KiCad** after installation through the package manager.
- **The last placer change** (mask openings as convex hulls) passed the test suite; set S was not rerun after it.
- **Micro vias and per-layer widths** have no fixture that needs them; they are tested for correctness on boards with the setting forced, not for benefit.
- **Three KiCad demo boards** still differ from KiCad's DRC in documented ways.
- **Global-routing set-up on LeeChee** takes 135 seconds because obstacle queries are about fifty times slower on that board; the cause was not found. Only the opt-in global options are affected.

11 Roadmap in full

Generated from dev/progress.json. A check mark is a closed task; its text says whether it was built or closed without building, and with which result.

M0: Toolchain and skeleton

Deliverables. CMake presets (debug, release, asan, cpu-only), CUDA build with a GCC host compiler that nvcc 12.4 accepts, Catch2, clang-format/clang-tidy config (or GCC equivalents), scripts/bootstrap.sh, kicad-cli available (Docker image or native), fixture download script

Gate. ctest runs a CPU and a CUDA hello-kernel test on both GPUs; kicad-cli version works from the harness

Tasks (14 of 14 closed; status: done).

- ✓ Ubuntu setup script (packages, Docker, KiCad 10 kicad-cli wrapper)
- ✓ Git repository, .gitignore
- ✓ CMake project + presets (debug, release, cpu-only, asan, tsan)
- ✓ CUDA build: nvcc 12.4 with g++-13 host, sm_60 + sm_70; system GCC 15 for C++
- ✓ Core library: units, deterministic Philox/SplitMix RNG
- ✓ GPU service: device discovery by UUID, free-memory admission control
- ✓ First kernel with CPU reference; bit-identical on V100 and P100
- ✓ Catch2 tests wired into ctest with JUnit results
- ✓ tracemaker CLI: version, gpu-info
- ✓ Development progress website (0.0.0.0)
- ✓ scripts/bootstrap.sh: configure, build and test all presets
- ✓ Benchmark environment check (kicad-cli from the harness)
- ✓ Fixture download script (Freerouting, DAC2020, KiCad demos, PCBench)
- ✓ Gate: CPU + CUDA tests pass on both GPUs; kicad-cli works from harness

M1: KiCad I/O and board model

Deliverables. s-expression lexer/parser/writer; .kicad_pcb, .kicad_pro, .kicad_dru, .kicad_sch (netlist), .kicad_mod; board model with layers, stackup, footprints, pads, tracks, vias, zones, nets, net classes; DSN reader

Gate. Byte-identical round-trip of every fixture board (parse → write); netlist equals `kicad-cli sch export netlist` on all fixture projects

Tasks (11 of 11 closed; status: done).

- ✓ Lossless s-expression parser/writer with span edits
- ✓ Board reader: layers, nets (KiCad 9 and 10 syntax), footprints, pads, tracks, arcs, vias, zones, graphics
- ✓ Reader verified against pcbnew ground truth on demo boards
- ✓ Byte-identical round trip of fixture boards
- ✓ Design rules: .kicad_pro net classes, patterns, assignments, board minimums
- ✓ Custom rules: .kicad_dru parsed into rules and constraints
- ✓ Board editor: add/remove tracks and vias, move/rotate footprints
- ✓ Edited boards load in pcbnew with the expected geometry
- ✓ Netlist import via kicad-cli (kicadsexpr) and cross-check against board pad nets
- ✓ tracemaker inspect: board summary and JSON dump
- ✓ Gate: round trip on all fixtures; reader matches KiCad

M2: Rules and DRC parity

Deliverables. Rule compiler (net classes, board constraints, custom rules subset), spatial indices, own DRC with KiCad violation names

Gate. Same violation multiset as `kicad-cli pcb drc` on all unrouted and routed fixture boards (documented, tested exceptions only)

Gate result. Result: 14 of 17 KiCad demo boards exact and 264 of 264 injected-defect variants (regression tests); three demo boards differ in documented ways. Since 2026-10-06 Margin-layer edges and one-sided SMD pads match KiCad too.

Tasks (11 of 11 closed; status: done).

- ✓ Exact integer geometry kernel (rounded-core shapes, int128 predicates, fine arc flattening)
- ✓ Copper item model: pads (all shapes, drill offsets), tracks, arcs, vias, zone fills, copper graphics with nets
- ✓ Rule engine: net classes, local overrides, diff pairs, board minimums, legacy rules, custom-rule conditions

- ✓ Checks: clearance, shorts, crossings, widths, vias, annular, holes, edge, keepouts
- ✓ Anchor/overlap connectivity, unconnected items, dangling tracks/vias
- ✓ DRC parity harness vs kicad-cli + fuzzer
- ✓ Parity: 11/16 demo boards exact (regression test)
- ✓ Parity on broken-board classification (shorts/dangling): 264/264 injected-defect variants match kicad-cli violation by violation (was 41/264; regression test); demo boards 14/17 exact
- ✓ Performance on very large boards: vme-wren 730 s -> 7 s, jetson 250 s -> 4 s (zone-fill edge index, area-rule cache; identical reports)
- ✓ Copper text obstacle margin (estimated glyph boxes)
- ✓ DRC parity 12 of 17 KiCad demo boards (jetson now exact; RoyalBlue54L down to a malformed duplicate pad)

M3: Event bus, server, viewer v1

Deliverables. Event types, ring buffers, WebSocket server, FlatBuffers schema, replay log; WebGL2 viewer (WebGPU backend optional): layers, pan/zoom, pads/tracks/vias/zones, ratsnest, dark theme

Gate. Viewer renders the largest fixture board at ≥ 60 fps; replay of a recorded log reproduces the final board

Gate result. Result: replay reproduces the routed board (test record_replay). The 60 fps figure was not measured: there is no GPU-backed browser on this machine.

Tasks (9 of 9 closed; status: done).

- ✓ Event types and Sink interface
- ✓ WebSocket server (Boost.Beast), static files, snapshot + deltas, back-pressure
- ✓ Viewer: WebGL2 instanced tracks/pads/vias/zones, layers, pan/zoom
- ✓ Live effects: glow, frontiers, failures, HUD, log, tooltips
- ✓ tracemaker-view with -demo
- ✓ Router live streaming (route -view)
- ✓ Replay log (zstd frames, -record FILE.zst) and heatmaps (expansions, history) live and in replay; test: recording/viewing leave the routed board byte-identical
- ✓ Event recording (route -record) and replay (tracemaker-view -replay); test: replay reproduces the routed board exactly
- ✓ FlatBuffers schema: closed, not built (protocol and replay log stay JSON, zstd-compressed logs 12-13x smaller; D45, D61)

M4: First router (lattice A*)

Deliverables. Rerun debug logging of searches (optional), transactions, scoring, union-find, bit-planes, A* + radix heap, insertion with exact validation, basic pull-tight; sequential routing in net order; **tracemaker route CLI**; bench harness v1 with KiCad-DRC judge and Freerouting runner

Gate. Routes the simple fixture set with 0 added KiCad DRC errors; harness produces the comparison table

Tasks (6 of 6 closed; status: done).

- ✓ Router obstacle model with exact KiCad-rule legality
- ✓ Octilinear lattice A* with vias, bends, admissible heuristic
- ✓ Path simplification, exact verification, commit through the board editor
- ✓ tracemaker route CLI
- ✓ Benchmark harness judged by kicad-cli with Freerouting baselines
- ✓ Gate: zero added KiCad DRC errors on the benchmark

M5: Failure memory T0–T2 + escalation R0–R3 + repair

Deliverables. Failure records, Zobrist signatures, nogood store, history + targeted penalties, conflict graph, activity ordering, bandit; negotiated rip-up; repair windows; aggressor/victim repair

Gate. Completion $\geq$ Freerouting on the Freerouting test set; ablation shows each learning mechanism helps or is removed

Tasks (10 of 10 closed; status: done).

- ✓ Negotiated rip-up and reroute (soft crossing of routed copper, victims requeued)
- ✓ PathFinder history on contested cells
- ✓ Learned blocked cells after exact-check failures
- ✓ Best-state keeping
- ✓ Failure records, nogoods, conflict graph, bandit (T0-T2 complete)
- ✓ Gate: completion $\geq$ Freerouting on tier A (100% clean, equal to Freerouting RC12, in every run since 2026-10-03)
- ✓ Reverse probe for boxed-in targets; pad legs dropped when the lattice end is on the pad; learned blocks beside fine-pitch pads
- ✓ Diversified extra restarts while budget remains
- ✓ Post-routing clean-up: via-saving re-routes and path smoothing (bends -40%, vias -25-33%)
- ✓ Connection planning by pad copper distance (–plan-gap, on, D54): 24 boards at a fixed budget 11,905 -> 11,986 routed; tiers A-D neutral; failure reasons now reported as at the best state

M6: Global routing + GPU

Deliverables. Port/adapt GAMER sweep and GGR pattern kernels from Xplace `gpugr` (BSD-3); tile graph, Steiner decomposition, CPU NCR, corridors; GPU pattern routing, GPU sweep maze routing, GPU cost-to-go fields, GPU DRC broad-phase

Gate. CPU and GPU results identical on every fixture; time-to-route reduced on dense boards with no quality loss

Tasks (10 of 10 closed; status: done).

- ✓ GPU cost-to-go fields (CUDA sweeps) with identical CPU reference
- ✓ Fields used as the router’s A* heuristic; deterministic with/without GPU
- ✓ Coarse global routing + corridors (CPU, negotiated congestion): built, off by default (corridors cost completion in three measurements)
- ✓ GPU pattern/maze global routing (GAMER/GGR port): closed, not built. The CPU global router routes the tile graph in 0.02-0.5 s per board and its plan does not help (D56)
- ✓ GPU DRC broad-phase: closed, not built. Broad-phase is 14 ms; the DRC was fixed on the CPU with a zone-fill edge index: vme-wren 730 s -> 7 s, jetson 250 s -> 4 s, reports byte-identical (D55)
- ✓ Adaptive GPU-field cut-off under GPU contention (wall-clock mode)
- ✓ CPU global router v1 (tile graph, capacities, negotiated congestion, soft corridors): experimental, no gain yet
- ✓ Global router v2: net-shared edges (Steiner demand), via capacity, layer-true corridors, integer costs, corridor-restricted windows: built; 18 hard boards 13,305 without, 12,967 corridors, 13,324 congestion map; off (D56)
- ✓ Exact flood-fill reachability pre-check before likely-failing searches (10 hard boards +47 routed at fixed budget; tiers neutral)
- ✓ Global congestion map + cut-based tile capacities (opt-in –global-congestion): 18 hard boards 13,285 -> 13,294 routed, within noise; kept off (D53). Boards are not congested at tile scale; v2 remainder deferred

M7: Placement A–D

Deliverables. Constraint extraction, decoupling detection, quadratic B2B (PCG), electrostatic global placement (CPU then GPU batched), rotation/side assignment, LP legalisation; placement report with optimality levels and lower bound

Gate. Placements are legal (zero courtyard overlaps, inside outline); $HPWL \leq SA\text{-}PCB$ baseline on the placement set

Gate result. Result: legality met (held-out set, doc 04 §10). The HPWL comparison with SA-PCB was NOT measured: the baseline does not build here (it needs SWIG, which is not installed, and its Makefile is stale; build/tools/sa-pcb).

Tasks (12 of 12 closed; status: done).

- ✓ Constraint extraction (fixed parts, outline, keepouts, courtyards, copper legality)
- ✓ Decoupling capacitors tied to their IC supply pin (D25)
- ✓ Other proximity groups: crystals, load caps, ESD near connectors, regulator caps (M13, –component-rules soft)
- ✓ quadratic B2B (Eigen CG)
- ✓ global spreading (SimPL instead of electrostatic; CPU)
- ✓ Rotation assignment (coordinate descent)
- ✓ Side assignment (flip to the back): opt-in –flip, annealer flip move + via estimate, KiCad-exact mirror (2,529 footprints = pcbnew); 23 boards HPWL -8%, crossings -31%, 0 new DRC (D48)
- ✓ legalisation (greedy minimum displacement; LP not done)
- ✓ parallel simulated annealing, deterministic across thread counts
- ✓ placement report with optimality levels and LP lower bound (min-cost flow)
- ✓ evaluation on 23 PCBench boards: HPWL -17..-26% median, zero new DRC errors; routability not yet improved
- ✓ –mode auto with route checks: 23 boards, HPWL -24.5% total, never more unrouted (61 -> 59)

M8: Placement E–G + ECO

Deliverables. GPU parallel-tempering SA, LNS, CP-SAT windows, routability loop; R4 exact window solve, R5 trial ordering, R6 cut proofs, ECO placement moves

Gate. Schematic-to-board set: clean pass $\geq 90\%$; eco mode closes boards that none mode cannot

Gate result. Result 2026-10-06: NOT met. From-scratch clean pass 85 % (34/40) on set S; 4 of the 6 failing boards cannot be clean in any placement. eco closed 0 of the 1 held-out board the human placement leaves unrouted (3 of 7 on the development set).

Tasks (12 of 12 closed; status: done).

- ✓ Parallel tempering (annealed replica exchange, CPU; deterministic for 1/3/16 threads): refine cost -4% on 23 boards, full neutral
- ✓ GPU parallel-tempering SA: closed, not built (annealing 1.9 s median, 56 s max on the held-out boards; CPU tempering neutral on routability; D57)
- ✓ LNS (window rip-up, exact ≤ 4 parts, greedy otherwise; never worsens)
- ✓ Exact windows by branch and bound (CP-SAT not available; B&B equals full enumeration)
- ✓ CP-SAT windows: closed, not built (no OR-tools C++ library; exact windows by branch and bound equal full enumeration; D57)
- ✓ RUDY routability term in the annealer (incremental, exact vs reference)
- ✓ Routability loop (–mode routable): 23 boards unrouted 64 -> 35 (auto 60), 0 new DRC errors
- ✓ ECO placement moves (–mode eco): closes 3 of 7 incomplete boards, unrouted 64 -> 48
- ✓ Schematic-to-board set S (40 held-out boards, parts piled) built and the gate measured: 85 % clean (34/40) with routable –scratch, human placement 95 %; gate of 90 % NOT met (doc 04 §10)
- ✓ R4 exact window solve: closed, not built as an exact model (no CP-SAT); the group re-route stand-in changed no board and was removed (doc 05 §17, §20)

- ✓ R5 trial ordering: closed, not built (every order of small conflict groups was tried and changed no board, doc 05 §17, §20)
- ✓ R6 cut proofs: built (route -cut-report); no over-full line on 92 tier B-D boards, tightest line 40 % used

M9: Escape planning

Deliverables. Min-cost-flow escape, layer assignment, NC fallback, escape templates

Gate. BGA boards in the academic set complete

Gate result. Result: NOT met. 17 dense-package boards: 41 % clean, 47 % of the 15 feasible ones (Freerouting 2.5: 24 %).

Tasks (9 of 9 closed; status: done).

- ✓ Escape corridors for dense-package pins (perimeter + dog-bone), opt-in -escape-plan
- ✓ Escape feasibility analysis: tracemaker escape, dead pins with reasons and hints; benchmark dead_pins / clean_pass_feasible
- ✓ Via neck-down rung (smallest via the board minimums allow)
- ✓ Dead pins: not retried after a negotiated neck-down check
- ✓ Second-ring channel corridors (built, off: measured mixed)
- ✓ Min-cost-flow channel assignment for deep arrays (built, opt-in -escape-flow: measured below the v1 corridors, D49)
- ✓ Per-ring layer assignment (per pin, layer by layer, inside -escape-flow; same evidence, D49)
- ✓ NC fallback for escapes: closed, not built (the flow planner it belongs to measured below the corridors and is off; ordinary negotiation already overrides corridors; D62)
- ✓ Escape templates in the knowledge base: closed, not built (planning takes 0.1-2 s per board; a cache saves nothing; D62)

M10: Portfolio, determinism, knowledge base

Deliverables. Successive-halving portfolio, deterministic sync points, persistent KB (T3), warm starts

Gate. Bit-identical output across thread counts and GPU on/off; clean pass up at fixed time budget

Tasks (10 of 10 closed; status: done).

- ✓ Parallel portfolio of 8 router variants, best kept
- ✓ Deterministic work budget (byte-identical repeat runs)
- ✓ Persistent knowledge base with variant bandit
- ✓ Restarts that keep history
- ✓ Successive-halving budget allocation (-halving, restart-based): 24 boards at 2 / 4 threads 12,385 / 12,451 routed against 12,402 / 12,508 for the first variants; kept off (D59)
- ✓ Bit-identical across thread counts (portfolio selection; -variants, all 8 with -work, D47)
- ✓ Portfolio early stop once a variant routes everything (wall-clock mode)
- ✓ 2x lattice pitch for two variants on large boards
- ✓ Quality benchmark vs Freerouting 2.5.0-RC12 and 1.9.0 on 60 held-out boards (bench/quality_bench.py)
- ✓ Held-out quality benchmark on tier C-like boards (heldC-1) and DAC2020

M11: KiCad integration

Deliverables. IPC plugin (live board in pcbnew, one undoable commit), Python bindings, PCM package

Gate. Round-trip in KiCad on the demo boards

Tasks (8 of 8 closed; status: done).

- ✓ IPC action plugin: save copy with project, route, add tracks/vias in one commit
- ✓ Offline test against kicad-python 0.8
- ✓ Round trip in a running KiCad GUI (headless: pcbnew in Docker under Xvfb, scripts/kicad_gui_roundtrip.py): tracks 140 -> 316 in the open board, one undo removes them, DRC of the saved board clean of new errors

- ✓ Python bindings (pybind11): read_board, route, emit_items, drc; GIL released; CLI parity test
- ✓ PCM package (scripts/make_pcm_package.py, validate_pcm.py; passes kicad-python 0.8 validator)
- ✓ Plugin uses the Python module when importable, binary otherwise
- ✓ Install the PCM archive in a running KiCad GUI (Plugin and Content Manager, Install from File: listed as installed and compatible; venv creation and toolbar button not checked, no network in the container)
- ✓ Plugin: footprint moves (TRACEMAKER_PLACE), re-routing of existing unlocked copper (TRACEMAKER_REROUTE, route -reroute), zone refill (TRACEMAKER_REFILL); checked in the running GUI

M12: Advanced rules and arms

Deliverables. Differential pairs, length/skew tuning, blind/micro vias, gridless tile-plane arm, learned ordering/congestion models

Gate. Each feature gated by its own benchmark subset; no regressions

Tasks (12 of 12 closed; status: done).

- ✓ Differential pairs: experimental -diff-pairs (rarely couples)
- ✓ Differential pairs: proper coupled routing (v2, opt-in -diff-pairs, D50): coupled pair search with coupled vias, breakout legs, re-coupling, enclosed-pin check, -pair-skew-mm; 18 PCBench boards: 35 of 59 pairs $\geq$ 50 % coupled at the rule gap, completion 2,201 -> 2,210; KiCad: no new error kinds (one more Margin-layer edge clearance on a board that has 27 with pairs off) (doc 05 §15)
- ✓ Differential pairs: twists (layer change with the sides swapped, on with pair routing: one of 59 pairs gains coupling, median share 57 -> 63 %), per-pair skew limits from component rules; pairs ending on routed copper measured to matter little (13 of 21 such attempts couple) and not built (D64)
- ✓ Length and skew tuning (KiCad length constraints)
- ✓ Blind/buried vias: experimental -blind-vias
- ✓ Micro vias (-micro-vias, off): outer layer to the next at the class's microvia size where a through via is blocked; no fixture allows them on more than two layers; KiCad-clean on two boards with the setting forced on
- ✓ Gridless tile-plane arm: closed, not built; a half-pitch portfolio variant for small boards takes its place (12 boards 4,140 -> 4,153 routed at half pitch; tier B 27 -> 28 clean; D63)
- ✓ Learned ordering/congestion models: closed, not built (ordering changes measured no effect, tile congestion absent, variant bandit already in the knowledge base; D64)
- ✓ Length tuning: custom length rules read; trombone meanders until in range; KiCad-judged test
- ✓ Differential pairs: coupled pair routing v1 (experimental, -diff-pairs; rarely succeeds without coupled escapes)
- ✓ Skew tuning for differential pairs (custom skew rules, inDiffPair condition)
- ✓ Blind/buried vias where through vias are blocked (-blind-vias; correct on the 3 boards that allow them, no measurable gain)

M13: Component-aware layout rules

Deliverables. Category detection (USB, Ethernet, crystals, regulators, RF, ...), role binding, rule catalogue compiled into net classes, a sidecar .kicad_dru, keep-outs and placer pseudo-nets; stackup impedance; report (doc 15)

Gate. Detection precision/recall on a labelled PCBench subset; every applied rule met or reported; no regression on the quick tier

Tasks (17 of 17 closed; status: done).

- ✓ Verify the 41 unchecked catalogue numbers against their sources (2026-10-05: 9 confirmed, 9 corrected, 11 TraceMaker defaults now D, 12 unverifiable stay R; doc 15 §8.10, D46)
- ✓ Category detection + role binding with confidence
- ✓ Rule compiler: sidecar .kicad_dru, keep-out areas, placer pseudo-nets
- ✓ Rule compiler: net classes from rules, used by the router with -component-rules on (tmk_<RULE>_<ref>, Default-class nets only, D58; doc 15 §15)

- ✓ Stackup parsing + impedance (IPC-2141A / Wadell) - widths/gaps reported per layer (P4, report only)
- ✓ Detection and rule report
- ✓ Labelled PCBench subset and benchmark
- ✓ Placer pseudo-nets for proximity rules (crystal, ESD, regulator caps, generalised decoupling): `-component-rules`
- ✓ Generated keep-outs (crystal, inductor) for router/DRC + sidecar `.kicad_dru`
- ✓ Per-net coupled routing for USB 2.0 pairs (`RouterOptions::pair_nets`)
- ✓ Net classes from rules: impedance width/gap from the stackup, IPC-2221 width for current, minimum widths (one width per net: front-layer solution; per-layer widths not built)
- ✓ IPC-2221 width for current reported for rules with a current (USBC-09)
- ✓ User override file (`-rules-override`, JSON): `disable` / `assert` / `deny` / `set`, validated, reported (doc 15 §6.3)
- ✓ Ethernet magnetics void (ETH-05) as a generated keep-out (`-component-rules` on)
- ✓ Connector edge attraction in the placer (`opt-in -edge-attraction`, objective only)
- ✓ Two-stage soft placement (decaps and their ICs locked first, D43)
- ✓ Turn component rules on by default for placement (soft; placed tiers B+C off vs soft: clean pass unchanged, crystals closer, decaps equal; D52). Routing stays off

A Every test

Release build, the full run that produced Table 5. Unit tests are grouped by source file with their Catch2 tags; integration tests carry their CTest labels and a description.

`src/crules/test_crules.cpp` (19 tests, 0.3 s)

Test	Tags / labels	Time (s)	Result
catalogue: embedded JSON loads with every category and rule	[crules]	0.01	passed
catalogue: schema errors are reported, not ignored	[crules]	0.00	passed
names: references, supplies, voltages, capacitances	[crules]	0.00	passed
detection and role binding on a synthetic board with pin names	[crules]	0.01	passed
detection is deterministic and independent of footprint order	[crules]	0.02	passed
boards without pin names: binding by nets and pad numbers, capped confidence	[crules]	0.01	passed
D25 ties are reproduced and the generalised decoupling is a superset	[crules]	0.00	passed
evaluation: proximity measured, statuses follow the mode	[crules]	0.01	passed
placement affinities: each part pulled by one rule, locked parts never	[crules]	0.02	passed
keep-outs: crystal area on the free layer, tracks only, named; sidecar rule	[crules]	0.01	passed
USB 2.0 pairs for coupled routing (USB2-02)	[crules]	0.01	passed
PCBench fixture: detection report is stable and finds the crystal	[crules][fixture]	0.05	passed
overrides: every form parses, in file order	[crules][overrides]	0.01	passed
overrides: typos and contradictions are errors with a clear message	[crules][overrides]	0.01	passed
overrides: disable a rule, a rule on one part, a category	[crules][overrides]	0.02	passed
overrides: deny and assert change which instances exist	[crules][overrides]	0.01	passed
overrides: set changes parameters; @REF wins over global, later wins among equals	[crules][overrides]	0.02	passed
ethernet: discrete magnetics bound and voided on the adjacent layer (ETH-05)	[crules][ethernet]	0.01	passed
ethernet: four layers, MagJack, through-hole magnetics	[crules][ethernet]	0.01	passed

`src/crules/test_impedance.cpp` (16 tests, 0.1 s)

Test	Tags / labels	Time (s)	Result
elliptic integral ratio $K(k)/K(k')$	[impedance]	0.00	passed
microstrip: Hammerstad & Jensen against textbook, KiCad and IPC-2141A	[impedance]	0.00	passed
microstrip: 50 ohm on 1.6 mm FR-4 is about 3 mm wide (doc 15 §5.3)	[impedance]	0.00	passed
stripline: Cohn/Wheeler against textbook and KiCad	[impedance]	0.00	passed
coupled microstrip: Kirschning & Jansen against KiCad	[impedance]	0.00	passed
coupled stripline: Cohn 1955 exact and against KiCad	[impedance]	0.00	passed
grounded coplanar waveguide: exact and against KiCad	[impedance]	0.00	passed
impedance is monotone where the solver relies on it	[impedance]	0.00	passed
solving: integer-nm bisection round trips and is deterministic	[impedance]	0.00	passed
propagation delay	[impedance]	0.00	passed
IPC-2221 width for current	[impedance][current]	0.00	passed
stackup geometry: microstrip outside, stripline inside, reference zones	[impedance][stackup]	0.00	passed
impedance rules: widths per layer with a stackup, never without one	[impedance][stackup]	0.01	passed
width for current from a rule's current, with the stackup's copper	[impedance][current]	0.01	passed
KiCad demo stackups parse and give sane geometry	[impedance][stackup][fixture]	0.04	passed
net classes from rules: computed widths become classes for Default-class nets only	[impedance][stackup][netclass]	0.01	passed

src/place/test_place.cpp (28 tests, 30.4 s)

Test	Tags / labels	Time (s)	Result
rot90 matches KiCad rotation	[place]	0.01	passed
convex hull	[place]	0.00	passed
power-like net names	[place]	0.00	passed
ground-like net names	[place]	0.00	passed
translated closer() agrees with geom::closer_than	[place]	0.01	passed
min-cost flow on a small graph	[place]	0.00	passed
HPWL lower bound is exact on a chain and bounds random placements	[place]	0.01	passed
raster fast path is conservative (raster free implies exactly legal)	[place]	0.12	passed
legalisation removes every overlap	[place]	0.64	passed
recording a placement never changes it	[place]	3.07	passed
full pipeline: legal, deterministic, incremental cost exact	[place]	6.95	passed
rotation descent is monotone	[place]	0.00	passed
extraction from a KiCad board	[place][fixture]	0.02	passed
decoupling capacitors are tied to an IC supply pin, objective only	[place][fixture]	0.01	passed
component-rule proximity pseudo-nets are objective only and skip D25-tied parts	[place][fixture][crules]	0.03	passed
edge pulls: one-pin anchored nets, exact incremental cost, objective only	[place][crules]	2.94	passed
edge pulls from component rules: movable connectors only, objective only	[place][fixture][crules]	0.03	passed
RUDY map: capacity inside the outline, scaling, incremental demand equals the reference	[place][m8]	1.09	passed
parallel tempering: legal, exact cost, identical for any thread count	[place][m8]	4.59	passed
LNS never worsens the cost and keeps the placement legal	[place][m8]	1.99	passed
exact window: branch and bound equals full enumeration and never worsens	[place][m8]	0.83	passed
ECO: only improving moves, locked parts never move, legal	[place][m8]	0.65	passed
P5: byte-identical board for 1, 3 and 16 threads (tempering + LNS + RUDY, refine)	[place][m8][fixture]	4.11	passed
routability loop: never worse than its best seed, locked parts fixed	[place][m8]	1.54	passed
flip: the placer's mirrored geometry is what the KiCad writer produces	[place][fixture][flip]	0.02	passed

Test	Tags / labels	Time (s)	Result
flip: legal, exact incremental cost with the via term, deterministic, only allowed parts flip	[place][fixture][flip]	1.09	passed
flip: with no part allowed to flip, -flip changes nothing	[place][flip]	0.62	passed
flip: swap states exchange absolute poses on either side	[place][flip]	0.00	passed

src/server/test_replay.cpp (5 tests, 5.2 s)

Test	Tags / labels	Time (s)	Result
replay log round-trips plain and zstd	[replay]	0.09	passed
replay log keeps empty lines, long lines and end_frame boundaries	[replay]	0.02	passed
a truncated compressed log yields its complete lines, a corrupt one throws	[replay]	0.08	passed
heat grid sizing, scaling and message	[heatmap]	0.01	passed
router heatmaps: emitted with a sink, routing unchanged by it	[heatmap][route]	5.00	passed

src/server/test_server.cpp (4 tests, 0.4 s)

Test	Tags / labels	Time (s)	Result
message builders put type first and classify	[server]	0.00	passed
board snapshot of pic_programmer has the board's objects	[server]	0.03	passed
viewer server serves files, streams snapshot then deltas, shuts down	[server]	0.01	passed
slow clients lose transient messages but not state	[server]	0.33	passed

tests/test_diff_pair.cpp (5 tests, 0.0 s)

Test	Tags / labels	Time (s)	Result
pair rule: tightest legal coupling without diff-pair settings	[diffpair]	0.00	passed
pair rule: net-class diff-pair width and gap, relaxed clearance only for KiCad pairs	[diffpair]	0.00	passed
pair rule: a custom diff_pair_gap rule wins, diff_pair_uncoupled limits the legs	[diffpair]	0.01	passed
pair geometry: offsets are symmetric and miters keep the offset distance	[diffpair]	0.00	passed
pair measurement: coupled share, gap and skew	[diffpair]	0.00	passed

tests/test_escape.cpp (5 tests, 3.1 s)

Test	Tags / labels	Time (s)	Result
escape plan: perimeter pins fan out, inner balls get one dog-bone site each	[escape]	0.00	passed
escape analysis: sbc's DRAM balls are blocked only by the solder-mask rule	[escape][fixture]	3.12	passed
escape flow: a 6 x 6 array escapes on its pad layer within every channel's capacity	[escape]	0.01	passed
escape flow: without channels between balls, inner rings take dog-bone vias and leave on the next layer	[escape]	0.01	passed
escape flow: shallow packages are planned exactly as version 1	[escape]	0.00	passed

tests/test_geom.cpp (7 tests, 0.6 s)

Test	Tags / labels	Time (s)	Result
rotation follows KiCad's convention and is exact for right angles	[geom]	0.00	passed
segment intersection handles crossing, touching and collinear cases	[geom]	0.00	passed
exact closer-than agrees with floating-point distance away from the boundary	[geom]	0.01	passed
shape gaps: round pads, tracks and polygons	[geom]	0.00	passed
arcs flatten within the error bound	[geom]	0.00	passed
polygon edge index answers disk tests exactly like closer_than	[geom]	0.08	passed
polygon edge index answers shape tests and gaps exactly like the linear reference	[geom]	0.52	passed

tests/test_gpu.cpp (4 tests, 2.0 s)

Test	Tags / labels	Time (s)	Result
CPU reference fill is deterministic	[gpu][cpu-reference]	0.00	passed
GPU Philox fill is bit-identical to the CPU reference on every device	[gpu][cuda]	0.68	passed
cost-to-go field: CPU reference is the exact shortest-path distance on a small case	[gpu][field]	0.01	passed
cost-to-go field: GPU equals CPU on random grids	[gpu][field][cuda]	1.30	passed

tests/test_kicad_io.cpp (8 tests, 0.4 s)

Test	Tags / labels	Time (s)	Result
demo boards round-trip byte for byte	[kicad][io]	0.14	passed
board reader matches KiCad's own model	[kicad][io]	0.16	passed
reading an edited document gives the edited board	[kicad][io]	0.03	passed
stackup: layers, sublayers, permittivity and copper mapping are read; text is untouched	[kicad][io][stackup]	0.00	passed
stackup: boards without one say so; missing permittivity makes the gap incomplete	[kicad][io][stackup]	0.01	passed
stackup: KiCad demo with dielectric sublayers round-trips	[kicad][io][stackup]	0.08	passed
flipping a footprint: KiCad's mirror, layers swapped, the rest byte for byte	[kicad][io][flip]	0.00	passed
flipping a KiCad 5 module: user copper names, signed arc sweeps	[kicad][io][flip]	0.00	passed

tests/test_netlist.cpp (1 tests, 0.0 s)

Test	Tags / labels	Time (s)	Result
schematic netlist agrees with the board's pad nets	[netlist][kicad]	0.01	passed

tests/test_rng.cpp (2 tests, 0.0 s)

Test	Tags / labels	Time (s)	Result
Philox4x32-10 matches the Random123 known-answer vectors	[core][rng]	0.00	passed
RNG streams are pure functions of (seed, stage, item, n)	[core][rng]	0.00	passed

tests/test_rules.cpp (4 tests, 0.0 s)

Test	Tags / labels	Time (s)	Result
lengths with units convert to nanometres	[rules]	0.00	passed

Test	Tags / labels	Time (s)	Result
net-class wildcard patterns	[rules]	0.00	passed
project net classes, patterns and minimums are read	[rules]	0.01	passed
classes inherit null values from Default; explicit assignments and custom rules are read	[rules]	0.01	passed

tests/test_sexpr.cpp (7 tests, 0.0 s)

Test	Tags / labels	Time (s)	Result
millimetre strings convert to nanometres exactly	[sexpr]	0.01	passed
documents parse into a navigable tree	[sexpr]	0.01	passed
unmodified documents write back byte for byte	[sexpr]	0.00	passed
malformed documents raise ParseError with a line number	[sexpr]	0.01	passed
edits remove, replace and append with KiCad indentation	[sexpr]	0.00	passed
append to an inline list moves the closing paren to its own line	[sexpr]	0.00	passed
replacing the same node twice keeps the last replacement	[sexpr]	0.00	passed

tests/test_units.cpp (1 tests, 0.0 s)

Test	Tags / labels	Time (s)	Result
millimetre <-> nanometre conversion is exact for KiCad values	[core][units]	0.00	passed

integration (tests/CMakeLists.txt) (16 tests, 261.6 s)

Test	Tags / labels	Time (s)	Result
kicad_edit_roundtrip Boards edited by TraceMaker (tracks and vias added, footprints moved and rotated) are read back by pcbnew in Docker; geometry must equal TraceMaker's own reading. Time limit 600 s.	kicad;integration	4.06	passed
kicad_drc_parity TraceMaker's DRC against <code>kicad-cli pcb drc</code> on 14 KiCad demo boards: the same violations, type by type (15 routing-relevant types). Time limit 1200 s.	kicad;integration	9.56	passed
kicad_drc_broken_parity 264 boards with injected defects (11 demo boards, 12 defect kinds, 2 seeds): TraceMaker's DRC must match KiCad violation by violation. Time limit 1800 s.	kicad;integration	35.91	passed
drc_zone_index DRC reports through the zone-fill edge index are byte-identical to the linear reference path (<code>drc --linear-zones</code>) on four zone boards. Time limit 1800 s.	integration	21.69	passed
global_route <code>route --global</code> and <code>--global-congestion</code> are repeatable byte for byte at a fixed work budget and stay within 5 connections of routing without them. Time limit 1800 s.	integration;route	15.98	passed
record_replay A board routed plainly, with <code>--record</code> and with <code>--view</code> is byte-identical; replaying the recorded log reproduces the routed board. Time limit 600 s.	integration	12.11	passed
python_bindings The Python module reads, routes, checks and emits items; <code>route</code> gives the same bytes as the CLI. Time limit 600 s.	python;integration	12.20	passed
pcm_package Builds the KiCad PCM archive and validates its layout and metadata (also with <code>kicad-python's</code> validator). Time limit 120 s.	kicad	0.18	passed

Test	Tags / labels	Time (s)	Result
length_tuning A net with a custom <code>length</code> rule is meandered into range; KiCad's DRC judges the result. Time limit 600 s.	kicad;integration	3.40	passed
crules_catalogue_sync The committed rule catalogue JSON equals <code>docs/component_rules.yaml</code> and the rule ids used in doc 15. Time limit 60 s.	crules	0.36	passed
crules_route Routing with <code>--component-rules on</code> : no new track enters a generated keep-out, the keep-outs are not written into the board, the sidecar <code>.kicad_dru</code> is written. Time limit 600 s.	crules;integration	6.64	passed
crules_dru_parses KiCad itself parses the generated sidecar <code>.kicad_dru</code> without errors. Time limit 900 s.	kicad;integration	5.02	passed
crules_two_stage Two-stage placement with component rules: the result is legal, and parts locked after stage 1 are where the decoupling-only placement put them. Time limit 3600 s.	integration	64.96	passed
reach_verify Every "unreachable" verdict of the flood-fill pre-check is re-checked by the full A*: zero mismatches. Time limit 600 s.	route;integration	12.64	passed
thread_determinism The same <code>--work</code> portfolio at 1, 3 and 8 threads (and with fewer variants) writes identical bytes. Time limit 3600 s.	route;integration	29.77	passed
diff_pair A USB hub's five differential pairs are routed coupled at the rule's gap, twice with identical bytes, and KiCad's DRC is clean. Time limit 3600 s.	route;kicad;integration	27.13	passed

B Final tier runs, board by board

Runs `final11-tierA` to `final11-tierD`: eight variants, 120 seconds, KiCad's DRC as judge. The last column is Freerouting 2.5.0-RC12's published result for the same board.

Tier A: 40 of 40 clean, 2637 of 2637 connections

Board	Conn.	Routed	Vias	Time (s)	Clean	Freerouting 2.5
655_testboard	57	57	33	29	yes	clean
atmel-programmer_atmel_programmer	39	39	5	8	yes	clean
breakout-boards_swd-and-uart	11	11	2	2	yes	clean
C-BISCUIT_buck-reg-5v	55	55	10	3	yes	clean
CubeSAT-Reaction-Wheel_Edison_Motor_Servo	8	8	0	0	yes	clean
custom_cpu--ALU_custom_cpu--ALU	83	83	22	68	yes	clean
Dekada_dekada_TopoR_curves	117	117	0	1	yes	clean
devtys0_IRis	52	52	7	7	yes	clean
esp-serial-terminal_esp-com	59	59	14	8	yes	clean
ESP8266-MQTT-battery-monitor-hw_battery-monitor	95	95	35	16	yes	clean
esp8266_envmonitor_environment-monitor-1.2	32	32	7	5	yes	clean
ESP_WiFiSwitch_WifiSwitch	22	22	5	5	yes	clean
fan_controller_fan_controller	115	115	33	39	yes	clean
Hangul-Clock_Hangul	209	209	43	54	yes	clean
Hardware_Playground_rpi_zero_ws2812	46	46	2	2	yes	clean
HES-V2__autosave-hes	36	36	0	2	yes	clean
Inkjet__autosave-InkjetDriver	65	65	33	33	yes	clean
kika-in-space_DS8500	44	44	8	1	yes	clean
kitspace_4_switch_array	24	24	0	2	yes	clean
kitspace_8_switch_array	54	54	0	2	yes	clean

Board	Conn.	Routed	Vias	Time (s)	Clean	Freerouting 2.5
kitspace_aquarius	171	171	88	120	yes	clean
kitspace_pmt_combiner	35	35	0	3	yes	clean
kitspace_Potentiometer_mount_4LED	8	8	0	1	yes	clean
kitspace_threeboard	119	119	35	30	yes	clean
komputer-klavier_KomputerKlavier	87	87	2	4	yes	clean
LogicBoxen_LogicBoxen	116	116	70	37	yes	clean
mearm-base-pcb_ServoPCB	14	14	1	2	yes	clean
mechkeys_1800-controller	166	166	41	40	yes	clean
mini-whip_mini-whip-power-feed	23	23	11	5	yes	clean
mosavr_pcie1x_backplane	110	110	11	29	yes	clean
opentilt_opentilt1	24	24	2	16	yes	clean
phone_rtty_interface_phone_rtty_rev_a	61	61	0	2	yes	clean
Photodiode_dethead	40	40	4	2	yes	clean
QRPCard_QRPCard	92	92	24	16	yes	clean
scimpv_volumebuffer	67	67	3	5	yes	clean
SmartLaserCO2-PCB__autosave-SmartLaserShield	107	107	22	44	yes	clean
spisolator_spisolator	31	31	12	3	yes	clean
TTNEnschedeMote_ArduinoNanoRN2483	52	52	9	3	yes	clean
uC3Moy_uC3Moy	41	41	1	8	yes	clean
WordClock_v2.0_WordClock_v2	50	50	12	3	yes	clean

Tier B: 28 of 40 clean, 5172 of 5380 connections

Board	Conn.	Routed	Vias	Time (s)	Clean	Freerouting 2.5
APM-RPi-Shield_APM-RPi-Shield	125	125	51	63	yes	clean
avr_ledprojector_avr_ledprojection-0402	176	175	86	120	no	clean
bumps_bumps	330	330	79	120	yes	clean
cat-trainer_tensy_base_pcb	67	67	4	24	yes	clean
digital_clock_led_clock_3_and_4_digit	253	253	43	81	yes	clean
espeverywhere_espeverywhere	84	84	38	48	yes	clean
EtherCAT_shield_v1_EtherCAT_shield_v1	272	244	128	120	no	12 open
fifogfx_c64cart	104	64	21	122	no	clean
gamelights_leds	30	30	10	1	yes	1 open
Hardware_Playground_E3_Pcb_rev0.3	341	321	245	121	no	38 open, 5 viol.
Hardware_Playground_serial_gw_ATMEGA328P	172	171	72	120	no	3 open
Hyper-RGB-W-LED-Controller_led_strip_controller	74	74	5	25	yes	clean
KiCad-Like-a-Pro-Tutorial_rf24-breakout-v1	10	10	2	5	yes	clean
kitspace_ardfpga	214	204	164	121	no	30 open
kitspace_BalthazarPSU3	204	204	50	24	yes	0 open, 30 viol.
kitspace_OtterPill	137	127	88	120	no	17 open
kitspace_sympetrum-v2%20FF1.1	146	146	139	77	yes	44 open, 18 viol.
kitspace_sympetrum-v2%20NFF1.1	115	115	29	29	yes	15 open
kitspace_tensy-fx	196	182	113	122	no	18 open
kitspace_test-jig	292	260	79	120	no	0 open, 1 viol.
kitspace_USBI2C01	73	73	25	20	yes	6 open
memsarray_mems_modules	10	9	2	16	no	3 open
MicroMaple_MicroMaple	111	101	120	120	no	2 open
motor-3xdrv8833-hw_ver1	195	195	136	120	yes	clean
NavigationThing_NavigationThing	90	90	7	4	yes	clean
ourglass__autosave-watch-v4	134	134	72	121	yes	clean
PCB_small_halogen_replacement	52	52	4	1	yes	4 open

Board	Conn.	Routed	Vias	Time (s)	Clean	Freerouting 2.5
poncho_fpga_fpga-dongle	413	372	292	120	no	78 open
ProtoHead_v4_Prototyped_v4_FAB_rc1	34	34	8	1	yes	0 open, 10 viol.
Prototyping_Workshop_Prototyping_PCB	80	80	7	13	yes	clean
Radio-FM-TEA5767_radio tea5767	83	83	3	2	yes	clean
retroreflectors_SALSAFLOCK	40	40	19	70	yes	3 open, 6 viol.
royer_royer	46	46	2	1	yes	clean
rxadc_14_rxadc_14	124	124	27	46	yes	1 open
scimpy_amp	90	90	14	32	yes	clean
TOBS_HybridChargeController	210	210	61	120	yes	clean
TPS63001-Breakout_TPS63001 Breakout 25x25	26	26	7	2	yes	clean
wavegen_wavegen	140	140	62	60	yes	9 open
WemosLoraTracker_wemosGPSrfm	53	53	23	37	yes	clean
yamsek.kicad_i2c_bridge_primary	34	34	14	4	yes	clean

Tier C: 19 of 30 clean, 9516 of 9825 connections

Board	Conn.	Routed	Vias	Time (s)	Clean	Freerouting 2.5
96boards-robomezzi_96boards-robomezzi	282	280	116	120	no	2 open
AmpOne_dev-AmpOne	321	321	73	120	yes	clean
Apple-M0110-BT_Apple M0110	224	224	34	67	yes	clean
Biscay_Blueeye_sipm	306	306	97	101	yes	clean
BUF-DiffToSE-ADAU1966_BUF-DiffToSE-ADAU1966	307	307	53	16	yes	clean
draco_draco	418	418	231	120	yes	0 open, 6 viol.
DronPi_dronPi	352	341	147	120	no	22 open, 2 viol.
ErgodoneWithHole_ErgoDone	356	354	41	120	no	19 open, 16 viol.
Hardware_Playground_multisensor_cr2032	205	205	60	84	yes	4 open
Hardware_Playground_sd_wav_pcm5100	396	395	83	120	no	6 open
Hardware_Playground_usb_shield	211	211	1	1	yes	clean
Horticulture_Light_EQ_3-Band_3 channel led driver rev0.4	368	368	47	38	yes	clean
jadonk_PocketBone	202	202	100	120	yes	15 open
kitspace_40-channel-hv-switching-board	604	604	455	122	yes	clean
kitspace_d20_tri_r0.3	553	412	288	120	no	347 open
kitspace_dropbot_control_board	307	307	117	93	yes	0 open, 1 viol.
kitspace_PSLab	339	339	184	121	yes	clean
mechkeys_58r-mxalps	361	357	250	121	no	29 open, 8 viol.
mechkeys_MF68+10	415	415	305	120	yes	13 open
Microdox-PCB_Microdox	223	223	21	76	yes	17 open
Microdox-PCB_Microdox-topMCU	216	216	15	49	yes	1 open
motor-high-current-sensors_CurrentSensorPCB_line	472	461	218	120	no	3 open
P8000_WDC_Emulator_P8000_WDC_Emulator	361	326	79	129	no	clean
Paperino_HW_paperino_breakout	11	11	0	0	yes	clean
PCIE-to-MXM-Adapter_PCIEx1toMXM3.0	188	158	59	121	no	clean
rvi_v2x_hardware_V2X	479	456	389	120	no	51 open
tiny-8088_Computer	284	284	348	120	yes	clean
TripleDelay2399_TripleDelay2399	256	256	5	8	yes	clean
wordclock_ledboard2	310	310	1	3	yes	clean
zx-sizif-512-ext_sizif512ext	498	449	321	121	no	147 open

Tier D: 12 of 22 clean, 16522 of 17043 connections

Board	Conn.	Routed	Vias	Time (s)	Clean	Freerouting 2.5
Aleste-520EX_aleste	1616	1376	895	120	no	446 open, 48 viol.
badge2016_Badge_init	76	76	27	5	yes	clean
CoreOne-xCORE200-Original_CoreOne	807	752	271	121	no	4 open
dorkyboard_keyboard	830	820	131	129	no	6 open
DSP-ADAU1452_DSP-ADAU1452	742	742	222	88	yes	38 open
EUC-VESC_electronics_sept	760	760	83	122	yes	1 open
front-end-modules_LimeSDR_Sony	814	810	214	121	no	clean
Hardware_Playground_heater_actuator_node	1039	1039	42	53	yes	clean
Horticulture_Light_EQ_3-Band_light	422	422	76	36	yes	clean
Keyboard_PCB_Keyboard	534	533	318	121	yes	clean
KiCad-Library_Teensy_test_layout	1268	1246	457	123	no	16 open, 79 viol.
LeeChee_1800	1160	1125	342	120	no	84 open
m2-electronics_m2fc	724	724	262	120	yes	clean
mechkeys_LFK78	893	862	680	122	no	97 open, 4 viol.
memsarray_mems_array	245	245	221	123	yes	clean
mini_mass_prog_bench_prog_rig	995	976	265	123	no	5 open
MonApollo_analog-board	1190	1112	640	121	no	9 open, 58 viol.
newer-motor-controllers_busparts	316	316	82	121	yes	0 open, 1 viol.
newer-motor-controllers_si31-3	316	316	80	124	yes	0 open, 1 viol.
opendous_Upconverter	813	813	45	20	yes	7 open, 8 viol.
perfplusplus_Ard-perf++	886	886	0	1	yes	clean
TCKB_Kicad_TCKB	597	571	232	122	no	44 open

C Every benchmark run on record

All runs with a `summary.json` under `bench/results`, oldest first. Clean and completion are per cent; “FR” is Freerouting 2.5.0-RC12’s clean pass on the same boards where published; “Err.” is the number of boards with an error added by routing.

Run	Set	Boards	Clean	Compl.	FR	Err.	Commit
m4-tierA-v1	PCBench tier A	40	70.0	91.0	100.0	1	
m5-tierA-v2	PCBench tier A	40	92.5	99.6	100.0	0	
m5-tierA-v3	PCBench tier A	40	82.5	99.5	100.0	0	
m5-tierB-v1	PCBench tier B	40	22.5	91.8	50.0	7	
fixcheck-1	PCBench tier A	7	14.3	85.6	42.9	4	
fixcheck-2	PCBench tier A	7	14.3	86.5	42.9	2	
fixcheck-3	PCBench tier A	2	0.0	88.4	0.0	0	
m5-tierA-v4	PCBench tier A	40	87.5	99.6	100.0	0	5b4980d
m5-tierB-v2	PCBench tier B	40	27.5	90.2	50.0	0	5b4980d
m6-tierB-portfolio	PCBench tier B	40	42.5	91.8	50.0	1	ae12d7a
fixcheck-4	PCBench tier A	1	0.0	95.9	0.0	0	ae12d7a
escapes-1	PCBench tier A	8	12.5	86.3	50.0	0	febb4b2
escapes-2	PCBench tier A	4	25.0	89.3	75.0	0	febb4b2
escapes-3	PCBench tier A	4	25.0	89.2	75.0	0	febb4b2
escapes-4	PCBench tier A	2	0.0	79.6	100.0	0	febb4b2
escapes-5	PCBench tier A	4	25.0	88.3	100.0	0	febb4b2
text-2	PCBench tier A	3	33.3	96.2	33.3	0	febb4b2
m6-tierA-v5	PCBench tier A	40	97.5	100.0	100.0	0	5cf61bd
m6-tierB-v3	PCBench tier B	40	45.0	92.8	50.0	0	5cf61bd
text-3	PCBench tier A	3	66.7	99.1	33.3	0	5cf61bd
m6-tierB-v4	PCBench tier B	40	42.5	93.6	50.0	1	0acd2dd
text-4	PCBench tier A	4	25.0	95.5	25.0	0	91b4256
m6-tierC-v1	PCBench tier C	30	40.0	95.7	46.7	5	0acd2dd
fixC-1	PCBench tier A	4	0.0	74.3	50.0	3	9c53fc1
fixC-2	PCBench tier A	3	0.0	77.5	33.3	3	9c53fc1
fixC-3	PCBench tier A	2	0.0	64.9	0.0	2	9c53fc1

Run	Set	Boards	Clean	Compl.	FR	Err.	Commit
neck-1	PCBench tier A	8	12.5	88.4	62.5	0	d513c01
m6-tierB-180s	PCBench tier B	40	45.0	94.0	50.0	0	9c53fc1
final-tierC	PCBench tier C	30	36.7	93.2	46.7	1	3f122f3
final-tierB	PCBench tier B	40	50.0	94.8	50.0	0	3f122f3
hort-fix	PCBench tier A	1	0.0	100.0	100.0	1	511aafa
hort-fix2	PCBench tier A	1	0.0	100.0	100.0	1	511aafa
ghost-1	PCBench tier A	2	0.0	90.4	100.0	1	f7a7af9
final2-tierC	PCBench tier C	30	33.3	92.8	46.7	1	511aafa
final2-tierB	PCBench tier B	40	52.5	94.8	50.0	0	511aafa
final3-tierA	PCBench tier A	40	100.0	100.0	100.0	0	c31d2c0
final3-tierB	PCBench tier B	40	55.0	95.1	50.0	0	c31d2c0
final3-tierC	PCBench tier C	30	46.7	93.5	46.7	0	c31d2c0
final4-tierB	PCBench tier B	40	60.0	95.8	50.0	0	1bd2f56
final4-tierC	PCBench tier C	30	50.0	96.9	46.7	0	1bd2f56
final5-tierB	PCBench tier B	40	60.0	95.8	50.0	0	7c8c0c0
final5-tierC	PCBench tier C	30	53.3	97.2	46.7	0	7c8c0c0
final5-tierA	PCBench tier A	40	100.0	100.0	100.0	0	124884e
final5-tierD	PCBench tier D	21	52.4	94.9	38.1	0	124884e
final6-tierB	PCBench tier B	40	60.0	96.2	50.0	1	3334970
final6-tierC	PCBench tier C	30	60.0	97.5	46.7	0	3334970
final7-tierA	PCBench tier A	40	100.0	100.0	100.0	0	047acd2
final7-tierB	PCBench tier B	40	62.5	96.1	50.0	0	047acd2
final7-tierC	PCBench tier C	30	56.7	97.5	46.7	0	047acd2
final7-tierD	PCBench tier D	21	52.4	95.3	38.1	0	c3bb838
final8-tierD	PCBench tier D	22	50.0	95.2	36.4	0	f26b43d
final8-tierB	PCBench tier B	40	65.0	96.1	50.0	0	f26b43d
final8-tierC	PCBench tier C	30	56.7	97.5	46.7	0	f26b43d
final8-tierA	PCBench tier A	40	100.0	100.0	100.0	0	f26b43d
cleanup-vc3-tierB	PCBench tier B	40	60.0	96.0	50.0	0	3424984
cleanup-vc3-tierC	PCBench tier C	30	53.3	97.2	46.7	0	3424984
cleanup-vc1-tierB	PCBench tier B	40	62.5	96.0	50.0	0	3424984
cleanup-vc1-tierC	PCBench tier C	30	53.3	97.3	46.7	0	3424984
final9-tierA	PCBench tier A	40	100.0	100.0	100.0	0	fa190cf
final9-tierD	PCBench tier D	22	54.5	92.9	36.4	0	fa190cf
final9-tierB	PCBench tier B	40	65.0	96.3	50.0	0	fa190cf
final9-tierC	PCBench tier C	30	60.0	97.4	46.7	0	fa190cf
parity-tierA	PCBench tier A	40	100.0	100.0	100.0	0	f350d9c
mask-tierA	PCBench tier A	40	100.0	100.0	100.0	0	5aa1e02
final10-tierD	PCBench tier D	22	50.0	95.0	36.4	0	a953ca0
final10-tierB	PCBench tier B	40	67.5	96.2	50.0	0	a953ca0
final10-tierC	PCBench tier C	30	60.0	97.4	46.7	0	a953ca0
bga-base	PCBench tier all	17	41.2	85.1	23.5	1	92441f6
place-smoke	PCBench tier A + placement (routable)	1	100.0	100.0	100.0	0	9f76df8
placed-tierA	PCBench tier A + placement (routable)	40	95.0	100.0	100.0	0	d1e8e07
placed2-tierA	PCBench tier A + placement (routable)	40	100.0	100.0	100.0	0	f9e4051
placed2-tierB	PCBench tier B + placement (routable)	40	62.5	96.4	50.0	0	f9e4051
placed2-tierC	PCBench tier C + placement (routable)	30	56.7	97.3	46.7	0	f9e4051
placed2-tierD	PCBench tier D + placement (routable)	22	54.5	95.0	36.4	0	f9e4051
placed3-tierD	PCBench tier D + placement (routable)	22	50.0	95.2	36.4	0	f6b4d42
placed3-tierB	PCBench tier B + placement (routable)	40	70.0	96.7	50.0	0	f6b4d42
placed3-tierC	PCBench tier C + placement (routable)	30	56.7	97.5	46.7	0	f6b4d42
bga-base2	PCBench tier A	17	41.2	85.2	23.5	0	f4ea483
bga-esc1	PCBench tier A	17	41.2	85.6	23.5	0	0a00414
m9cur-tierA	PCBench tier A	40	100.0	100.0	100.0	0	34aa05a
m9esc-tierA	PCBench tier A	40	100.0	100.0	100.0	0	34aa05a
m9v-tierA	PCBench tier A	40	100.0	100.0	100.0	0	1956b69
m9ve-tierA	PCBench tier A	40	100.0	100.0	100.0	0	1956b69
m9ve-tierB	PCBench tier B	40	67.5	96.7	50.0	0	1956b69
m9v-tierB	PCBench tier B	40	67.5	96.6	50.0	0	1956b69
m9v-tierC	PCBench tier C	30	63.3	97.5	46.7	0	b734330
m9ve-tierC	PCBench tier C	30	63.3	97.4	46.7	0	b734330
m9p-tierB	PCBench tier B	40	67.5	96.6	50.0	0	52409bf
m9p-tierC	PCBench tier C	30	63.3	97.5	46.7	0	52409bf
m9f-tierA	PCBench tier A	40	100.0	100.0	100.0	0	0bb2778
m9f-tierD	PCBench tier D	22	50.0	96.2	36.4	0	0bb2778
m9p2-tierB	PCBench tier B	40	67.5	96.6	50.0	0	bd4806f
m9p2-tierC	PCBench tier C	30	63.3	97.5	46.7	0	bd4806f

Run	Set	Boards	Clean	Compl.	FR	Err.	Commit
final-tierA	PCBench tier A	40	100.0	100.0	100.0	0	70c832f
heldC-m9	PCBench tier A	30	50.0	97.1	43.3	0	a0a0df0
cap1m-tierB	PCBench tier B	40	67.5	96.6	50.0	0	822a362
cap1m-tierC	PCBench tier C	30	60.0	97.5	46.7	0	822a362
reach1-tierB	PCBench tier B	40	67.5	96.7	50.0	0	b084b14
reach1-tierC	PCBench tier C	30	63.3	97.5	46.7	0	b084b14
reach1-tierA	PCBench tier A	40	100.0	100.0	100.0	0	b084b14
reach1-tierD	PCBench tier D	22	50.0	96.2	36.4	0	b084b14
crules-off-tierB	PCBench tier B + placement (routable, component rules off)	40	70.0	97.0	50.0	0	7a8edaf
crules-soft-tierB	PCBench tier B + placement (routable, component rules soft)	40	70.0	96.9	50.0	0	7a8edaf
crules-soft-tierC	PCBench tier C + placement (routable, component rules soft)	30	60.0	97.7	46.7	0	d9b4a10
crules-off-tierC	PCBench tier C + placement (routable, component rules off)	30	60.0	97.7	46.7	0	d9b4a10
merged-tierA	PCBench tier A	40	100.0	100.0	100.0	0	3e6d2df
m12-tierA	PCBench tier A	40	100.0	100.0	100.0	0	88f53a2
plangap-tierD	PCBench tier D	22	54.5	95.8	36.4	0	179f138
plangap-tierB	PCBench tier B	40	67.5	96.6	50.0	0	179f138
plangap-tierC	PCBench tier C	30	63.3	97.5	46.7	0	179f138
ab-gap-tierD	PCBench tier D	22	54.5	96.2	36.4	0	179f138
ab-nogap-tierD	PCBench tier D	22	54.5	96.1	36.4	0	179f138
plangap-tierA	PCBench tier A	40	100.0	100.0	100.0	0	179f138
m6close-tierA	PCBench tier A	40	100.0	100.0	100.0	0	695e3f8
H-none-1	H (held-out, human placement)	40	95.0	99.9	85.0	1	2aea96b
S-full-1	S (from scratch) + placement (full)	40	70.0	95.6	85.0	1	2aea96b
S-full-2	S (from scratch), failing 12 + placement (full)	12	0.0	88.0	75.0	1	2aea96b
S-full-3	S (from scratch) + placement (full)	40	70.0	96.5	85.0	1	2aea96b
S-full-4	S (from scratch), failing 12 + placement (full)	12	25.0	90.8	75.0	1	2aea96b
S-routable-4	S (from scratch), failing 12, routable + placement (routable)	12	41.7	77.7	75.0	1	2aea96b
S-routable-5	S (from scratch), routable + placement (routable)	40	82.5	97.0	85.0	0	2aea96b
S-routable-6	S (from scratch), failing 7, long loop + placement (routable)	7	14.3	82.3	71.4	0	2aea96b
H-none-2	H (held-out, human placement)	40	95.0	99.9	85.0	0	2aea96b
S-routable-7	S (from scratch), routable + placement (routable)	40	85.0	96.9	85.0	0	2aea96b
H-eco-1	H boards not clean in the human placement, eco + placement (eco)	2	50.0	98.4	50.0	0	8634019
S-two-c0.6	S two boards, courtyard clearance 0.6 + placement (routable)	2	0.0	96.7	50.0	0	38b88eb
S-two-c1.0	S two boards, courtyard clearance 1.0 + placement (routable)	2	0.0	64.4	50.0	0	38b88eb
ab-nofine-tierB	PCBench tier B	40	67.5	96.5	50.0	0	b401162
ab-nofine-tierC	PCBench tier C	30	60.0	97.4	46.7	0	b401162
ab-fine-tierB	PCBench tier B	40	70.0	97.2	50.0	0	b401162
ab-fine-tierC	PCBench tier C	30	60.0	97.4	46.7	0	b401162
final11-tierA	PCBench tier A	40	100.0	100.0	100.0	0	78dad0a
final11-tierD	PCBench tier D	22	54.5	95.3	36.4	0	78dad0a
final11-tierB	PCBench tier B	40	70.0	97.2	50.0	0	78dad0a
final11-tierC	PCBench tier C	30	63.3	97.5	46.7	0	78dad0a